Conic Sections

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In ΔABC studying degrees, radians, and angles as rotations, $m\angle A = 35^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 93°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 52^\circ) = 93^\circ$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

For the model $f(x) = 3x + 3$ in trigonometric functions in the coordinate plane, evaluate $f(2)$.

Final Answer: 9. Substitute $x = 2$: $f(2) = 3(2) + 3 = 9$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A data set recorded for trigonometric functions in the coordinate plane consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 18.5. The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

An ellipse has vertices at $(1, 6)$ and $(1, -4)$, and a focus at $(1, 4)$. Determine its standard equation and compute its eccentricity.

The vertices share $x = 1$, so the major axis is vertical. Center is the midpoint of the vertices: $(h, k) = \left(1, \frac{6 + (-4)}{2}\right) = (1, 1)$. Semi-major axis: $a = 6 - 1 = 5$ (so $a^2 = 25$). Focal distance from center to $(1, 4)$: $c = 4 - 1 = 3$. For an ellipse, $b^2 = a^2 - c^2 = 25 - 9 = 16 \Rightarrow b = 4$. Standard equation: $\frac{(x-1)^2}{16} + \frac{(y-1)^2}{25} = 1$. Eccentricity: $e = \frac{c}{a} = \frac{3}{5} = 0.6$.

5 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the vertex, focus, and directrix of the parabola given by the general equation $y^2 + 8x - 4y + 28 = 0$.

Complete the square on the y-terms: $y^2 - 4y = -8x - 28 \Rightarrow (y^2 - 4y + 4) = -8x - 28 + 4$.
$(y - 2)^2 = -8x - 24 = -8(x + 3)$. This matches $(y - k)^2 = 4p(x - h)$ with:Vertex: $(h, k) = (-3, 2)$
Focal parameter: $4p = -8 \Rightarrow p = -2$ (opens horizontally to the left)Focus:
$(h + p, k) = (-3 + (-2), 2) = (-5, 2)$Directrix: $x = h - p = -3 - (-2) \Rightarrow x = -1$

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Conic Sections & Analytic Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Identify the conic section $4x^2 - 9y^2 - 16x - 18y - 29 = 0$ by completing the square. Find its center, vertices, foci, and asymptotes.

Step 1: Group and complete the square: $4(x^2 - 4x + 4) - 9(y^2 + 2y + 1) = 29 + 16 - 9 = 36$. $\mathbf{\frac{(x-2)^2}{9} - \frac{(y+1)^2}{4} = 1}$. Step 2: Extract hyperbola properties ($a = 3, b = 2, c = \sqrt{9+4} = \sqrt{13}$):Center: $(2, -1)$Vertices: $(-1, -1)$ and $(5, -1)$Foci: $(2 \pm \sqrt{13}, -1)$Asymptotes: $y + 1 = \pm\frac{2}{3}(x - 2)$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Eliminate the parameter t from the parametric curve $x(t) = 3 + 4\cos t$, $y(t) = -2 + 5\sin t$ for $0 \le t \le 2\pi$. Identify the conic section, its center, and its vertices.

Step 1: Isolate $\cos t$ and $\sin t$: $\cos t = \frac{x-3}{4}$, $\sin t = \frac{y+2}{5}$. Step 2: Apply the Pythagorean identity $\cos^2 t + \sin^2 t = 1$: $\mathbf{\frac{(x-3)^2}{16} + \frac{(y+2)^2}{25} = 1}$. This is a vertical ellipse centered at $\mathbf{(3, -2)}$ with major-axis vertices at $\mathbf{(3, -7)}$ and $\mathbf{(3, 3)}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

Equate the two equations: $2\sin\theta = 2\cos\theta \Rightarrow \tan\theta = 1$. On $[0, 2\pi)$, $\theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$.For $\theta = \frac{\pi}{4}$: $r = 2\sin\left(\frac{\pi}{4}\right) = \sqrt{2}$, giving point $\left(\sqrt{2}, \frac{\pi}{4}\right)$.For $\theta = \frac{5\pi}{4}$: $r = 2\sin\left(\frac{5\pi}{4}\right) = -\sqrt{2}$. However, $\left(-\sqrt{2}, \frac{5\pi}{4}\right)$ represents the identical geometric point as $\left(\sqrt{2}, \frac{\pi}{4}\right)$.Next, check the pole.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Decompose the vector $\mathbf{u} = \langle 6, 2\rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4\rangle$.

Step 1: Compute dot product and squared magnitude: $\mathbf{u} \cdot \mathbf{v} = (6)(3) + (2)(4) = 18 + 8 = 26$. $||\mathbf{v}||^2 = 3^2 + 4^2 = 9 + 16 = 25$. Step 2: Vector projection (parallel component): $\mathbf{u}_{\parallel} = \text{proj}_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v} = \frac{26}{25}\langle 3, 4\rangle = \left\langle \frac{78}{25}, \frac{104}{25}\right\rangle = \langle 3.12, 4.16\rangle$. Step 3: Orthogonal component.

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11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Evaluate the exact value of $\cos\left(\arcsin\left(-\frac{3}{5}\right)+\arctan(2)\right)$.

Let $\alpha=\arcsin\left(-\frac{3}{5}\right)$ and $\beta=\arctan(2)$. By definition: $\alpha\in[-\pi/2,0]$ (Quadrant IV), $\sin\alpha=-\frac{3}{5}\Rightarrow\cos\alpha=\sqrt{1-(-3/5)^2}=\frac{4}{5}$. $\beta\in(0,\pi/2)$ (Quadrant I), $\tan\beta=\frac{2}{1}\Rightarrow\text{hypotenuse}=\sqrt{1^2+2^2}=\sqrt{5}$, so $\sin\beta=\frac{2}{\sqrt{5}}$, $\cos\beta=\frac{1}{\sqrt{5}}$. Apply the cosine addition formula $\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 3 · AdvancedFind the matrix R that rotates vectors by $\theta=\frac{2\pi}{3}$ radians counterclockwise, and use it to find the image of the vector $\vec{w}=(4;\ -2)$. Verify that the magnitude of $\vec{w}$ is preserved under the transformation.

Step 1 (Formulate Rotation Matrix): For $\theta=\frac{2\pi}{3}$: $\cos(2\pi/3)=-\frac{1}{2}$ and $\sin(2\pi/3)=\frac{\sqrt{3}}{2}$: $R=\left(-1/2,\ -\sqrt{3}/2;\sqrt{3}/2,\ -1/2\right)$. Step 2 (Transform $\vec{w}$): $R\vec{w}=(-1/2,\ -\sqrt{3}/2;\sqrt{3}/2,\ -1/2)(4;\ -2)=((-1/2)(4)+(-\sqrt{3}/2)(-2);(\sqrt{3}/2)(4)+(-1/2)(-2))=(-2+\sqrt{3};2\sqrt{3}+1)$. Step 3 (Verify Magnitude Preservation): $|\vec{w}|^2=4^2+(-2)^2=16+4=20$. .

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the points $A(1,2,3)$, $B(2,4,1)$, and $C(-1,0,5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing A, B, C, and write the Cartesian equation of the plane.

Step 1: Form vectors $\vec{AB}$ and $\vec{AC}$: $\vec{AB}=\langle 2-1,4-2,1-3\rangle=\langle 1,2,\ -2\rangle$. $\vec{AC}=\langle -1-1,0-2,5-3\rangle=\langle -2,\ -2,2\rangle$. Step 2: Compute normal vector via cross product: $\mathbf{n}=\vec{AB}\times\vec{AC}=|\mathbf{i},\mathbf{j},\mathbf{k};1,2,\ -2;\ -2,\ -2,2|=\mathbf{i}(4-4)-\mathbf{j}(2-4)+\mathbf{k}(-2-(-4))=0\mathbf{i}+2\mathbf{j}+2\mathbf{k}=\langle 0,2,2\rangle$. Simplifying by scalar.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

For what value of k will the system have no solution?

$\{3x-2y=8;6x+ky=5$ Solution: 1. A linear system has no solution if the lines are parallel (identical slopes, different y-intercepts). 2. Express the first line in slope-intercept form: $-2y=-3x+8\Rightarrow y=\frac{3}{2}x-4\quad\Rightarrow m_1=\frac{3}{2}$. 3. Express the second line in slope-intercept form: $ky=-6x+5\Rightarrow y=-\frac{6}{k}x+\frac{5}{k}\quad\Rightarrow m_2=-\frac{6}{k}$. 4. Equate slopes to make lines parallel.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the exact value of $\cos\left(\frac{7\pi}{12}\right)$ and $\sin\left(\frac{7\pi}{12}\right)$ using angle addition identities.

Step 1: Decompose $\frac{7\pi}{12} = \frac{\pi}{3} + \frac{\pi}{4}$: Step 2: Apply cosine and sine sum formulas: $\cos\left(\frac{7\pi}{12}\right) = \cos\frac{\pi}{3}\cos\frac{\pi}{4} - \sin\frac{\pi}{3}\sin\frac{\pi}{4} = \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) - \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}-\sqrt{6}}{4}$. .

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Polynomial Functions

Prove by mathematical induction that for all integers $n \geq 1$: $1 + 3 + 5 + \ldots + (2n-1) = n^2$.

1. Base Step ($n = 1$): LHS $= 2(1) - 1 = 1$, RHS $= 1^2 = 1$. Since LHS = RHS, the base case $P(1)$ is true. 2. Inductive Hypothesis: Assume $P(k)$ is true for some integer $k \geq 1$: $1 + 3 + 5 + \ldots + (2k-1) = k^2$. 3. Inductive Step ($n = k+1$): Show that $1 + \ldots + (2k-1) + (2(k+1)-1) = (k+1)^2$: LHS $= [1 + 3 + \ldots + (2k-1)] + (2k+1)$. By the inductive hypothesis, substitute k^2.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Rational Expressions

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

Substitute $\cos(2\theta) = 1 - 2\sin^2\theta$ to express the entire equation in terms of $\sin\theta$: $(1 - 2\sin^2\theta) + 3\sin\theta = 2$. $-2\sin^2\theta + 3\sin\theta - 1 = 0 \Rightarrow 2\sin^2\theta - 3\sin\theta + 1 = 0$. Factor the quadratic in $\sin\theta$: $(2\sin\theta - 1)(\sin\theta - 1) = 0$. Case 1: $2\sin\theta - 1 = 0 \Rightarrow \sin\theta = \frac{1}{2}$. On $[0, 2\pi)$, $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$. Case 2: $\sin\theta - 1 = 0 \Rightarrow \sin\theta = 1$. On.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle θ between them to the nearest tenth of a degree.

Calculate the dot product: $\mathbf{a} \cdot \mathbf{b} = (2)(-3) + (-5)(4) = -6 - 20 = -26$. Calculate the magnitudes: $|\mathbf{a}| = \sqrt{2^2 + (-5)^2} = \sqrt{4 + 25} = \sqrt{29} \approx 5.385$, $|\mathbf{b}| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Scalar projection of $\mathbf{a}$ onto $\mathbf{b}$: $\text{comp}_{\mathbf{b}}\mathbf{a} = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{b}|} = \frac{-26}{5} = -5.2$. Angle θ between vectors.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Find the area enclosed between $y = x^2$ and $y = x + 2$.

Step 1 (Intersection points): Set $x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x-2)(x+1) = 0$. The curves intersect at $x = -1$ and $x = 2$. Step 2 (Definite Integral): On $[-1, 2]$, the line $y = x + 2$ is upper and the parabola $y = x^2$ is lower: $A = \int_{-1}^{2}((x+2) - x^2)\,dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2}$ Step 3 (Evaluate): At $x = 2$: $2 + 4 - \frac{8}{3} = \frac{10}{3}$. At $x = -1$: $\frac{1}{2} - 2 + \frac{1}{3} = -\frac{7}{6}$..

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Sequences & Series

Find all solutions to $2\cos^2\theta - \cos\theta - 1 = 0$ on the interval $[0, 2\pi)$.

Step 1: Factor the quadratic equation in $\cos\theta$: $(2\cos\theta + 1)(\cos\theta - 1) = 0$ Step 2: Set each factor to zero: Case 1: $2\cos\theta + 1 = 0 \Rightarrow \cos\theta = -\frac{1}{2}$. Cosine is negative in Quadrants II and III with reference angle $\frac{\pi}{3}$: $\theta = \pi - \frac{\pi}{3} = \mathbf{\frac{2\pi}{3}}$, $\theta = \pi + \frac{\pi}{3} = \mathbf{\frac{4\pi}{3}}$ Case 2: $\cos\theta - 1 = 0 \Rightarrow \cos\theta = 1$. On $[0, 2\pi)$, $\cos\theta = 1$ at $\theta = \mathbf{0}$. Final.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying trigonometric functions as right triangle ratios, $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 102° ✓

b. 112°

c. 94°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 45^\circ) = 102^\circ$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying special right triangles and exact value trigonometric ratios, $m\angle A = 34^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 101° ✓

b. 111°

c. 93°

d. 146°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (34^\circ + 45^\circ) = 101^\circ$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for trigonometric functions on the unit circle consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 4x + 4$ in trigonometric functions on the unit circle, evaluate $f(4)$.

a. 20 ✓

b. 24

c. 16

d. 24

Substitute $x = 4$: $f(4) = 4(4) + 4 = 20$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for evaluating trigonometric functions: exact values consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 5$ in evaluating trigonometric functions: exact values, evaluate $f(4)$.

a. 25 ✓

b. 30

c. 20

d. 29

Substitute $x = 4$: $f(4) = 5(4) + 5 = 25$.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the equation of the circle with center $(3, -1)$ and radius 5.

a. $(x + 3)^2 + (y - 1)^2 = 25$

b. $(x - 3)^2 + (y + 1)^2 = 25$ ✓

c. $(x - 3)^2 + (y - 1)^2 = 5$

d. $x^2 + y^2 = 25$

Applying Conic Sections principles, Standard circle equation. Therefore, the verified correct option is $(x - 3)^2 + (y + 1)^2 = 25$.

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find where the circle $(x-3)^2+(y-4)^2=25$ crosses the x-axis.

a. $(3,0)$ only

b. $(-3,0)$ and $(6,0)$

c. No intersection

d. $(0,0)$ **and** $(6,0)$ ✓

Set $y=0$: $(x-3)^2+16=25 \Rightarrow (x-3)^2=9 \Rightarrow x=0,6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A circle $\mathcal{C}$ with center $C(0,k)$ on the positive y-axis passes through the focus F of the parabola $x^2=4y$. If $\mathcal{C}$ is tangent to the parabola at two distinct points, P and Q, what is the area of quadrilateral $PFQC$?

a. $8\sqrt{3}$ ✓

b. $4\sqrt{3}$

c. $16\sqrt{3}$

d. $6\sqrt{3}$

1. **Identify the focus and circle equation:** The standard form of the parabola is $x^2=4py$, so with $4p=4$, we have $p=1$. Thus, the focus is $F(0,1)$. The circle is centered at $C(0,k)$ with $k>1$ and passes through $F(0,1)$, so its radius is $R=k-1$. The equation of $\mathcal{C}$ is: $x^2+(y-k)^2=(k-1)^2$ 2. **Find the tangency condition:** Substitute $x^2=4y$ from the parabola into the circle's equation: $4y+y^2-2ky+k^2=k^2-2k+1$ $y^2-2(k-2)y+(2k-1)=0$ For.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The equation of an ellipse is given by: $9x^2+4y^2-36x+8y+k=0$ If the length of the major axis of the ellipse is 6, what is the value of the constant k?

a. 4 ✓

b. -31

c. -41

d. -104

To determine k, we rewrite the equation in standard form by completing the square for both x and y: 1. Group the x and y terms and isolate the constant k: $9(x^2-4x)+4(y^2+2y)=-k$ 2. Complete the square inside each set of parentheses. Remember to distribute the leading coefficients when adding terms to the right-hand side: $9(x^2-4x+4)+4(y^2+2y+1)=-k+9(4)+4(1)$ $9(x-2)^2+4(y+1)^2=40-k$ 3. Divide both sides by $40-k$ (assuming $40-k>0$).

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An ellipse $\mathcal{E}$ and a hyperbola $\mathcal{H}$ are confocal, sharing the foci $F_1(-4, 0)$ and $F_2(4, 0)$. The eccentricity of $\mathcal{E}$ is $\frac{4}{5}$, and the eccentricity of $\mathcal{H}$ is 2. If P is an intersection point of $\mathcal{E}$ and $\mathcal{H}$ in the first quadrant, what is the area of ΔPF_1F_2?

a. $6\sqrt{3}$ ✓

b. $12\sqrt{3}$

c. $3\sqrt{3}$

d. $4\sqrt{5}$

We can solve this using either the geometric focal definitions or algebraic coordinate equations. **Method 1: Focal Distance Properties** The foci are located at $(\pm c, 0)$ where $c = 4$, so the base of ΔPF_1F_2 along the x-axis has length $|F_1F_2| = 2c = 8$. 1. For ellipse $\mathcal{E}$ with eccentricity $e_E = \frac{c}{a_E} = \frac{4}{5}$: $a_E = \frac{c}{e_E} = \frac{4}{4/5} = 5$ By the focal definition of an ellipse, any point P on $\mathcal{E}$ satisfies: $|PF_1| + |PF_2| = 2a_E = 10$ 2. For hy.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A parabola has its vertex at the origin $(0, 0)$ and passes through the point $(2, -4)$. If the directrix of the parabola is a horizontal line, what is the equation of the directrix?

a. **y = 4** ✓

b. y = -4

c. x = 4

d. x = -4

The vertex of the parabola is at $(0, 0)$. Since the directrix is a horizontal line, the axis of symmetry is vertical, and the equation of the parabola is of the form $x^2 = 4py$. The parabola passes through the point $(2, -4)$. Substituting these coordinates into the equation, we get $2^2 = 4p(-4)$, which simplifies to $4 = -16p$. Solving for p, we find $p = \frac{4}{-16} = -\frac{1}{4}$. The equation of the parabola is $x^2 = 4(-\frac{1}{4})y$, or $x^2 = -y$. The directrix of a parabola of the form $x^2 = 4py$ i.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A circle has its center at the point $(2, -3)$ and a radius of 5 units. Which of the following is the standard form equation of this circle?

a. $(x - 2)^2 + (y + 3)^2 = 25$ ✓

b. $(x + 2)^2 + (y - 3)^2 = 5$

c. $(x - 2)^2 + (y + 3)^2 = 5$

d. $(x + 2)^2 + (y - 3)^2 = 25$

The standard form equation of a circle with center (h, k) and radius r is given by $(x - h)^2 + (y - k)^2 = r^2$. In this problem, the center is $(h, k) = (2, -3)$ and the radius is $r = 5$. Substituting these values into the standard form equation, we get $(x - 2)^2 + (y - (-3))^2 = 5^2$, which simplifies to $(x - 2)^2 + (y + 3)^2 = 25$.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Derive the reflection property: light from one focus of an ellipse reflects to the other.

a. **By Snell's law variant: angle of incidence = angle of reflection using the definition $PF_1 + PF_2 = 2a$ and tangent bisects angle F_1PF_2** ✓

b. Only for circles

c. Only for parabolas

d. Arbitrary property

Applying Conic Sections principles, Optical property of ellipse. Therefore, the verified correct option is By Snell's law variant: angle of incidence = angle of reflection using the definition $PF_1 + PF_2 = 2a$ and tangent bisects angle F_1PF_2.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Prove that the chord joining $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ on $y^2 = 4ax$ has equation $y(t_1 + t_2) = 2x + 2at_1t_2$.

a. Use calculus

b. Incorrect formula

c. **Slope $= 2/(t_1 + t_2)$; using point-slope through $(at_1^2, 2at_1)$ gives the result** ✓

d. Only for $t_1 = t_2$

Applying Conic Sections principles, Algebraic derivation. Therefore, the verified correct option is Slope $= 2/(t_1 + t_2)$; using point-slope through $(at_1^2, 2at_1)$ gives the result.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the equation of the hyperbola with asymptotes $y = \pm 2x$ and passing through $(3, 4)$.

a. $\frac{x^2}{5} - \frac{y^2}{20} = 1$ ✓

b. $y^2 - 4x^2 = 0$

c. $4x^2 - y^2 = 32$

d. $x^2 - 4y^2 = 4$

A hyperbola with asymptotes $y = \pm 2x$ has the form $4x^2 - y^2 = k$. Substituting $(3, 4)$ gives $4(9) - 16 = 20$, so $4x^2 - y^2 = 20 \Rightarrow \frac{x^2}{5} - \frac{y^2}{20} = 1$.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A chord PQ of the parabola $y^2 = 4ax$ passes through the focus. Prove that $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$ where S is the focus.

a. Not always true

b. **Using focal chord parametrization $t_1 t_2 = -1$ and $SP = a(1 + t_1^2)$** ✓

c. Only for the latus rectum

d. Use AM-GM

Applying Conic Sections principles, Classic focal chord property. Therefore, the verified correct option is Using focal chord parametrization $t_1 t_2 = -1$ and $SP = a(1 + t_1^2)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Derive the eccentricity of a conic from the focus-directrix ratio definition $PF/PD = e$ for the parabola.

a. $e = 2$ for parabola

b. **For parabola, $e = 1$: distance to focus equals distance to directrix** ✓

c. $e > 1$ for parabola

d. $e < 1$ always

Applying Conic Sections principles, Definition of parabola. Therefore, the verified correct option is For parabola, $e = 1$: distance to focus equals distance to directrix.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Find the values of k so that the line $y = kx - 1$ is tangent to the parabola $x^2 = 4y$.

a. $k = 0$

b. $k = \pm 1$ ✓

c. Any k

d. $k = 2$

Substituting $y = kx - 1$ into $x^2 = 4y$ gives $x^2 - 4kx + 4 = 0$. For tangency, the discriminant is zero: $(-4k)^2 - 16 = 0 \Rightarrow 16k^2 = 16 \Rightarrow k = \pm 1$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Sequences & Series

Prove that the difference of the focal radii of a point on a hyperbola equals $\pm 2a$.

a. **From definition $|PF_1 - PF_2| = 2a$, which defines the hyperbola** ✓

b. Only for $e > 2$

c. $PF_1 - PF_2 = 2c$

d. It equals $2b$

Applying Conic Sections principles, Definition of hyperbola. Therefore, the verified correct option is From definition $|PF_1 - PF_2| = 2a$, which defines the hyperbola.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Trigonometric Functions

For the ellipse $b^2x^2 + a^2y^2 = a^2b^2$, confirm this equals standard form.

a. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ✓

b. $x^2 + y^2 = a^2 + b^2$

c. $ax + by = ab$

d. $b^2 + a^2 = 1$

Applying Conic Sections principles, Divide both sides by a^2b^2. Therefore, the verified correct option is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Functions & Their Graphs

Find the polar equation of an ellipse with focus at the origin, semi-major $a = 5$, $e = 0.6$.

a. $r = \frac{5}{1 - \cos\theta}$

b. $r = \frac{5}{1 + 0.6\cos\theta}$

c. $r = \frac{3.2}{1 - 0.6\cos\theta}$ ✓

d. $r = \frac{8}{1 + 0.6\cos\theta}$

For an ellipse with focus at the pole, $r = \frac{a(1-e^2)}{1 - e\cos\theta} = \frac{5(1-0.36)}{1 - 0.6\cos\theta} = \frac{3.2}{1 - 0.6\cos\theta}$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Matrices & Systems

Convert the hyperbola $4x^2 - 9y^2 = 36$ to standard form.

a. $\frac{x^2}{9} - \frac{y^2}{4} = 1$ ✓

b. $x^2 - y^2 = 9$

c. $\frac{x^2}{36} - \frac{y^2}{36} = 1$

d. $\frac{x^2}{4} - \frac{y^2}{9} = 1$

Applying Conic Sections principles, Divide by 36: $x^2/9 - y^2/4 = 1$. Therefore, the verified correct option is $\frac{x^2}{9} - \frac{y^2}{4} = 1$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Probability & Statistics

Find the equation of the ellipse with foci $(\pm 3, 0)$ and sum of focal radii $= 10$.

a. $\frac{x^2}{10} + \frac{y^2}{9} = 1$

b. $\frac{x^2}{9} + \frac{y^2}{16} = 1$

c. $\frac{x^2}{25} + \frac{y^2}{16} = 1$ ✓

d. $x^2/16 + y^2/9 = 1$

Applying Conic Sections principles, $2a = 10$, $a = 5$. $b^2 = 25 - 9 = 16$. Therefore, the verified correct option is $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the equation of the circle with diameter endpoints $(2, 3)$ and $(8, 7)$.

a. $(x-5)^2 + (y-5)^2 = 13$ ✓

b. $(x-5)^2 + (y-5)^2 = 26$

c. $(x-5)^2 + (y-5)^2 = 25$

d. $x^2 + y^2 = 13$

Applying Conic Sections principles, Center $(5, 5)$, $r^2 = (3^2 + 2^2) = 13$. Therefore, the verified correct option is $(x-5)^2 + (y-5)^2 = 13$.

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